Query Analysis Report

Objective:

To extract meaningful insights from the datasets using structured SQL queries, each aligned with a specific business goal. Queries were designed to answer both strategic and tactical questions from multiple perspectives. The goal was to answer business-critical questions across four core themes:

1. Business Questions
2. Customer Behavior
3. Customer Feedback
4. Sales Insight

All queries were executed in Google BigQuery, using structured, documented SQL logic. Each query is designed to drive specific visuals or KPIs in the final dashboard.

Tools & Environment:

1. **Database**: Google BigQuery
2. **Query Language**: SQL
3. **Editor**: BigQuery Console
4. **Source Tables:**
    - dim_date
    - fact_sales
    - dim_customers
    - dim_travel_package
    - dim_travel_guide
    - dim_feedback
    - dim_refund

SQL Files Overview:

File Name	**Description**
dim_date.sql	Date dimension table creation.
Business Questions.sql	High-level business questions & KPIs
Customer Behavior Insight Framework.sql	Insights on customer type, age, loyalty
Customer Feedback Analysis.sql	Sentiment, recommendation, and guide performance
Sales Insight Framework.sql	Trends in revenue, bookings, discounts

Analysis:

1. Business Questions

Business Problem: Adress management-level KPI and strategic decisions.

#	Question	SQL Query	Insights Summary
1.	Top 3 Travel Packages by Monthly Revenue.	SELECT * FROM `ppnj.dim_date`; WITH monthly_revenue AS (SELECT d.month month, d.year year, tp.package_name, SUM(s.num_of_pax * tp.price_per_pax - s.discount_promotion) total_revenue FROM `ppnj.dim_date` d JOIN `ppnj.fact_sales` s ON d.date_key = s.travel_date JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id GROUP BY d.year, d. month, tp.package_name), ranked_revenue AS (SELECT *, RANK() OVER(PARTITION BY package_name ORDER BY total_revenue DESC) package_rank FROM monthly_revenue) SELECT * FROM ranked_revenue WHERE package_rank <=3;	- Nha Trang emerged as the top revenue generator, reaching RM 33, 736 in October 2023. - Vientiane appeared 3 times in the Top 3 across May, July, and November 2023, with a consistent revenue range around RM 23,477 to RM 28,597. - Jakarta made the Top 3 in June, July, and September 2023, indicating a strong mid-year performance. - Putrajaya remained competitive in August 2023 and January 2024, showing potential in domestic travel offerings. # Visuals Recommendation: - Line Chart (small multiples)
2.	Average Discount per Channel and Usage Rate	SELECT channel, COUNT(*) total_bookings, ROUND(AVG(discount_promotion), 2) avg_discount, SUM(CASE WHEN discount_promotion > 0 THEN 1 ELSE 0 END) discounted_bookings, ROUND(SUM(CASE WHEN discount_promotion > 0 THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) dicount_usage_rate FROM `ppnj.fact_sales` GROUP BY channel;	- Walk-in customers received the highest average discount of RM 456.09, and the highest discount usage rate 61.98%, despite having fewest total bookings of 121. - Across all channels, discount usage rates were relatively consistent around 60%, suggesting uniform promotion strategies.
3.	High-Value One-Time Customers	WITH customer_spending AS (SELECT c.customer_id, c.customer_name, c.customer_type, c.age, COUNT(s.booking_id) AS total_bookings, SUM(s.num_of_pax * tp.price_per_pax - s.discount_promotion) AS total_spent FROM `ppnj.fact_sales` s JOIN `ppnj.dim_customers` c ON s.customer_id = c.customer_id JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id GROUP BY c.customer_id, c.customer_name, c.customer_type, c.age) SELECT * FROM customer_spending WHERE total_bookings = 1 ORDER BY total_spent DESC;	- All are categorized as regular customers, with ages ranging between 18 and 40 years. - The highest spender with RM 38, 116, only booked once and did not return. - A few customers represent a younger high-value that may be worth targeting in retention campaigns # Visual Recommendation: Highlight table with customer name, age, total spent, and a visual tag for customer type. Optional: Use a funnel or retention chart showing drop-off after 1st high-value trip.

4.	Top 5 Guides with Lowest Ratings	SELECT g.staff_name, ROUND(AVG(fb.rating_guide), 2) AS avg_guide_rating, COUNT(DISTINCT r.refund_id) AS total_refunds, COUNT(fs.booking_id) AS total_bookings FROM \`ppnj.fact_sales\` fs JOIN \`ppnj.dim_travel_guide\` g ON fs.guide_id = g.staff_id LEFT JOIN \`ppnj.dim_feedback\` fb ON fs.booking_id = fb.booking_id LEFT JOIN \`ppnj.dim_refund\` r ON fs.booking_id = r.booking_id GROUP BY g.staff_name ORDER BY avg_guide_rating ASC LIMIT 5;	- The bottom 5 guides received average ratings between 3.13 and 3.55, which is significantly lower than the expected service standard. - These results highlight potential service quality concerns, customer dissatisfaction, or mismatches in customer expectations. # Visual Recommendation - A horizontal bar chart comparing average guide ratings. - A heatmap table with conditional formatting to flag guides with both low ratings and high refund counts.

5.	Refund Rate Among Low Satisfaction Customers	WITH low_feedback AS (
		 SELECT booking_id
		 FROM \`ppnj.dim_feedback\`
		 WHERE satisfaction_score <= 3
		),
		refund_stats AS (
		 SELECT
		  COUNT(*) AS low_feedback_count,
		  SUM(CASE WHEN r.booking_id IS NOT NULL THEN 1 ELSE 0 END) AS refunded_count
		 FROM low_feedback lf
		 LEFT JOIN \`ppnj.dim_refund\` r ON lf.booking_id = r.booking_id
		)
		SELECT
		 low_feedback_count,
		 refunded_count,
		 ROUND(refunded_count * 100.0 / low_feedback_count, 2) AS refund_rate
		FROM refund_stats;
			- A total of 28 bookings were flagged with low satisfaction scores.
			
			- None of these bookings resulted in a refund request.
			
			- The refund rate among dissatisfied customers is 0.0%.
			
			# Visual Recommendation
			- A KPI card showing “0.0% refund rate among dissatisfied customers”.
			- Supplement with a donut chart visualizing low-satisfaction customers by refund outcome.

6.	Top 3 Month of Peak revenue	WITH revenue_by_month AS (SELECT d.month month, d.monthname monthname, COUNT(s.booking_id) total_bookings, SUM(s.num_of_pax * tp.price_per_pax - s.discount_promotion) total_gross_revenue FROM \`ppnj.fact_sales\` s JOIN \`ppnj.dim_date\` d ON s.booking_date = d.date_key JOIN \`ppnj.dim_travel_package\` tp ON s.package_id = tp.package_id GROUP BY d.month, d.monthname), rank_month AS (SELECT *, RANK() OVER(ORDER BY total_gross_revenue DESC) month_rank FROM revenue_by_month) SELECT * FROM rank_month WHERE month_rank <= 3;	- August is the top-performing month, generating the highest gross revenue despite not having the highest number of bookings. - June and July follow closely in revenue, showing strong performance in the mid-year period. - This suggests that Q3 (July–September) is the peak travel season, possibly influenced by school holidays, festive periods, or marketing efforts. # Visual Recommendation - Bar chart showing gross revenue per month (highlighting top 3). - Optionally, add a dual-axis line chart showing both revenue and number of bookings for better context.

2. Customer Behaviour Insight Framework

Business Problem: Discover customer characteristics and booking habits.

#	Question	SQL Query	Insights Summary
1.	Customer Demographic Profile	SELECT gender, customer_type, COUNT(customer_id) total_customers, ROUND(AVG(age),0) avg_age FROM `ppnj.dim_customers` GROUP BY gender, customer_type;	- The customer base is evenly distributed between males and females, with a slight majority of regular female customers (420). - The average age across all segments falls within a narrow range (34–36 years), indicating a consistent age profile of customers. - The "Other" gender category has relatively low representation (55 customers total), but with age patterns consistent with the other groups.
2.	Preferred Package Name by Traveller type	SELECT tp.package_name, tp.package_type, COUNT(s.booking_id) total_booking FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id JOIN `ppnj.dim_customers` c ON s.customer_id = c.customer_id GROUP BY tp.package_type, tp.package_name;	- Family and honeymoon packages dominate in booking volume. - Traveller types exhibit clear destination preferences, opening opportunities for custom itineraries or bundled deals. - Solo travellers lean toward accessible cities, while honeymooners seek beach and island packages.

3.	Repeat vs. New Customers	SELECT
		 CASE
		  WHEN booking_count > 1 THEN 'repeat'
		  ELSE 'new'
		 END customer_status,
		 COUNT(*) num_customers
		FROM (
		 SELECT
		  customer_id,
		  COUNT(*) booking_count
		 FROM `ppnj.fact_sales`
		 GROUP BY customer_id
		) sub
		GROUP BY customer_status;
			- A majority (57.7%) of the customer base are new customers, suggesting strong top-of-funnel acquisition.
			
			- However, a healthy 42.3% have returned for multiple bookings, indicating positive customer experience and brand trust.
			
			- The split also shows potential to grow loyalty further by nurturing first-time customers into repeat clients.

4.	Booking Lead Time by Customer Type	SELECT tp.package_type, ROUND(AVG(DATE_DIFF(s.travel_date, s.booking_date, DAY)), 0) avg_lead_days FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id GROUP BY tp.package_type;	- Across all package types, customers book approximately 6 weeks in advance. - Solo travelers book slightly earlier than others — an average of 46 days in advance — possibly reflecting more flexibility and early planning behavior. - Family, group, and honeymoon travelers book 43–44 days ahead, likely due to the need for coordination among multiple people or event timing (e.g., honeymoons).

5.	Top 1 preference by age group for each destination	WITH age_group_bookings AS (SELECT CASE WHEN c.age < 25 THEN 'under-25' WHEN c.age BETWEEN 25 AND 34 THEN '25-34' WHEN c.age BETWEEN 35 AND 49 THEN '35-49' ELSE '50+' END age_group, tp.destination, COUNT(s.booking_id) total_booking FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id JOIN `ppnj.dim_customers` c ON s.customer_id = c.customer_id GROUP BY age_group, tp.destination), ranked_booking AS (SELECT *, ROW_NUMBER() OVER(PARTITION BY destination ORDER BY total_booking DESC) rank_destination FROM age_group_bookings) SELECT age_group, destination, total_booking FROM ranked_booking WHERE rank_destination = 1;	- The 35–49 age group is the dominant traveler segment for the majority of destinations. - The 25–34 age group leads bookings for some popular and adventurous locations, such as: • Cameron Highlands (21 bookings) • Langkawi (14) • Pulau Redang (13) • Ho Chi Minh City (13) • Yogyakarta (19) - The under-25 and 50+ age groups did not emerge as top travelers for any destination in this analysis, suggesting they may be smaller market segments. - Travelers aged 35–49 appear to be the primary audience, especially for destinations like: • Borneo, Dili, and Bangkok - Destinations favored by 25–34-year-olds tend to be nature-based or offbeat urban escapes. # Visual Recommendations: - Heatmap or matrix chart: Age group vs. Destination with booking volume. - Bar chart showing top destinations for 25–34 and 35–49 age groups side by side.
6.	Customer Channel Preferences	WITH age_group_booking AS (SELECT CASE WHEN c.age < 25 THEN 'under-25' WHEN c.age BETWEEN 25 AND 34 THEN '25-34' WHEN c.age BETWEEN 35 AND 49 THEN '35-49' ELSE '50+' END age_group, s.channel, COUNT(s.booking_id) total_booking FROM `ppnj.fact_sales` s JOIN `ppnj.dim_customers` c ON s.customer_id = c.customer_id GROUP BY s.channel, age_group), rank_by_age_group AS (SELECT *, ROW_NUMBER() OVER(PARTITION BY age_group ORDER BY total_booking DESC) rn FROM age_group_booking) SELECT age_group, channel, total_booking FROM rank_by_age_group WHERE rn = 1;	- All age groups — including seniors (50+) — predominantly use the online channel for bookings. - The 35–49 group is the most active online, followed by 25–34. - Online is the dominant booking channel across all age segments. - Despite assumptions that older generations may prefer offline methods, the 50+ group also prefers online bookings, though at a lower volume. # Visual Recommendations: - Matrix chart: Age group vs. channel with booking volume. - Stacked bar chart: Total bookings segmented by age group and channel.

3. Customer Feedback Analysis

Business Problem: Understand satisfaction, loyalty, and refund behavior.

#	Questions	SQL Query	Insights Summary
1.	Average Satisfaction Score by Package	SELECT tp.package_name, COUNT(f.feedback_id) num_feedback, ROUND(AVG(f.satisfaction_score), 2) avg_score FROM `ppnj.dim_travel_package` tp JOIN `ppnj.dim_feedback` f ON tp.package_id = f.package_id GROUP BY tp.package_name ORDER BY avg_score DESC;	- The highest-rated package is Yangon with an average score of 8.33, suggesting very strong customer satisfaction. - Pulau Sipadan and Vientiane also deliver exceptional customer experiences. - At the other end, Ho Chi Minh City, Phuket, and Dili have the lowest scores, indicating a need for improvement.
2.	Ratings Breakdown by Travel Guide	SELECT tg.staff_name, ROUND(AVG(f.rating_guide),2) avg_guide_rating, COUNT(f.feedback_id) total_reviews FROM `ppnj.dim_travel_guide` tg JOIN `ppnj.fact_sales` s ON s.guide_id = tg.staff_id JOIN `ppnj.dim_feedback` f ON s.booking_id = f.booking_id GROUP BY tg.staff_name ORDER BY avg_guide_rating DESC;	- Nasir is the highest-rated guide with an average of 3.77, followed closely by Nasrul and Alissa. - Intan has the lowest rating at 3.13, which may indicate a need for performance review or customer engagement improvement. - Most guide ratings hover between 3.1 to 3.8, indicating overall moderate satisfaction, but with room for improvement.
3.	Would Recommend Rate	SELECT ROUND(SUM(CASE WHEN recommend_group = 'Yes' THEN num_feedback ELSE 0 END) * 100.0 / SUM(num_feedback), 2) AS yes_percentage FROM (SELECT CASE WHEN would_recommend THEN 'Yes' ELSE 'No' END AS recommend_group, COUNT(*) AS num_feedback FROM `ppnj.dim_feedback` GROUP BY recommend_group);	- Nearly 8 out of 10 customers (79.38%) would recommend the company's services to others. - This suggests a high level of satisfaction, likely driven by good experiences with packages, customer service, or pricing. - However, the remaining ~21% detractors are still significant and warrant investigation into common pain points. # Visual Recommendation: - Donut Chart: Display percentage split between "Yes" and "No".
4.	Refund Rate by Package	SELECT p.package_name, COUNT(r.refund_id) AS total_refunds, COUNT(fs.booking_id) AS total_bookings, ROUND(COUNT(r.refund_id) * 100.0 / COUNT(fs.booking_id), 2) AS refund_rate FROM `ppnj.fact_sales` fs JOIN `ppnj.dim_travel_package` p ON fs.package_id = p.package_id LEFT JOIN `ppnj.dim_refund` r ON fs.booking_id = r.booking_id GROUP BY p.package_name ORDER BY refund_rate DESC;	- Bagan tops the list with a 12% refund rate, a red flag for service quality or experience issues. - Manila and Krabi follow closely with 8.7% each, warranting immediate attention. - Several packages such as Chiang Mai, Langkawi, and Melaka have moderate refund rates between 5% to 7%. - Encouragingly, over 40% of packages (17 out of 39) have zero refunds, indicating strong operational and customer satisfaction performance.
5.	Refund Amount & Reason for refund	SELECT refund_status, COUNT(*) AS num_refunds, SUM(refund_amount) AS total_refund_value, ROUND(AVG(refund_percentage), 2) AS avg_percent FROM `ppnj.dim_refund` GROUP BY refund_status;	- Approved refunds dominate both in count and value, making up over 55% of total refunds. - Even pending and rejected refunds carry significant amounts, signalling unresolved issues that still impact customer sentiment.

4. Sales Insight Framework

Business Problem: Explore trends in bookings, revenue, and promotions.

#	Questions	SQL Query	Insights Summary
1.	Total Bookings & Revenue Over Time	SELECT d.year, d.monthname, COUNT(fs.booking_id) total_booking, SUM(fs.num_of_pax * tp.price_per_pax - fs.discount_promotion) total_revenue, ROUND(AVG(fs.num_of_pax * tp.price_per_pax - fs.discount_promotion), 2) avg_revenue_per_booking FROM `ppnj.fact_sales` fs JOIN `ppnj.dim_date` d ON fs.booking_date = d.date_key JOIN `ppnj.dim_travel_package` tp ON fs.package_id = tp.package_id GROUP BY d.year, d.monthname ORDER BY d.year, d.monthname;	- August recorded the highest total revenue at RM 1,155,382, driven by 138 bookings. - July had the highest number of bookings at 147, but slightly lower revenue (RM 1,063,219) than August — possibly due to cheaper packages or more discounts applied. - March and November stood out with the highest average revenue per booking at RM 9,453.30 and RM 9,560.54 respectively, despite lower booking volumes. - May to July saw consistent growth in bookings, showing strong mid-year performance.
2.	Top Performing Travel Packages	SELECT tp.package_name, COUNT(s.booking_id) total_booking, SUM(s.num_of_pax * tp.price_per_pax) total_revenue FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id GROUP BY tp.package_name ORDER BY total_revenue DESC;	- Hanoi leads with RM 1,120,230 in total revenue, generated from 40 bookings — suggesting either high package pricing, larger group sizes, or both. - Phnom Penh and Krabi made it into the top 5 with fewer bookings (34 and 23, respectively), implying these are high-value packages likely popular with premium or international travellers. - Cameron Highlands had the most bookings (50) but appears lower on revenue rank (RM 133,680) — suggesting it's a budget-friendly or domestic favorite package.
3.	Channel Performance Analysis	SELECT s.channel, COUNT(s.booking_id) total_booking, SUM(tp.price_per_pax * s.num_of_pax) total_revenue FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id GROUP BY s.channel ORDER BY total_revenue DESC;	- The online channel clearly dominates with 613 bookings and generating RM 5.18 million in total revenue. - This reflects a strong customer preference for digital booking platforms and possibly effective online marketing strategies.
4.	Discount Impact on Revenue	SELECT CASE WHEN s.discount_promotion > 0 THEN 'discounted' ELSE 'non-disounted' END AS discount_group, COUNT(s.booking_id) AS bookings, SUM(tp.price_per_pax * s.num_of_pax) gross_revenue, SUM(discount_promotion) total_discounted, SUM(tp.price_per_pax * s.num_of_pax - s.discount_promotion) net_revenue FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id GROUP BY discount_group;	- Discounted bookings account for 748 bookings, generating a gross revenue of RM 6.3 million. - After applying RM 457.95k in discounts, the net revenue is still strong at RM 5.85 million. - This suggests that discounts successfully attract higher sales volume, compensating for the reduced margins. - non-discounted bookings are fewer (486 bookings), they still generated RM 4.12 million, with zero revenue loss from promotions. - Discounted bookings contribute 58% of total bookings and 59% of net revenue, showing a balanced impact.

5.	Revenue by Travel Month	SELECT d.monthname, SUM(s.num_of_pax * tp.price_per_pax - s.discount_promotion) revenue FROM `ppnj.fact_sales` s JOIN `ppnj.dim_travel_package` tp ON s.package_id = tp.package_id JOIN `ppnj.dim_date` d ON s.travel_date = d.date_key GROUP BY d.month, d.monthname ORDER BY d.month ASC;	- With RM 1.15 million in revenue, September generated the highest monthly travel revenue, indicating a strong seasonal demand—possibly due to school holidays or local events - June (RM 907k), July (RM 886k), and August (RM 920k) also showed high revenue, reinforcing the mid-year period as a prime travel season for targeted promotions. - February (RM 572k) and March (RM 715k) recorded the lowest revenues, suggesting off-peak months with lower demand—ideal for running aggressive marketing campaigns or flash deals. - November and December maintained healthy revenue levels, possibly linked to year-end holidays and Christmas break, though not as strong as mid-year or September.

Techniques Used:

- Window Functions: RANK(), ROW_NUMBER()
- CTEs: For breakdown of logic and reusability
- Aggregations: SUM(), AVG(), COUNT()
- CASE Statements: For conditional groupings
- JOINs: Across dimensions and fact tables
- QA Checks: For null or placeholder values

Output Usage:

Query results power visuals in Power BI, including:

- Revenue & booking trends
- Loyalty & churn charts
- Guide performance metrics
- Refund dashboards
- Segment-specific insights

Summary Conclusion:

- Structured SQL queries were used to answer strategic and operational business questions across four themes: Business Questions, Customer Behavior, Customer Feedback, and Sales Insight.
- Analysis revealed Q3 (July–September) as the peak travel season, with August and September showing the highest revenue.
- Online booking is the dominant channel across all age groups, contributing over RM 5 million in total revenue.
- Identified high-value one-time customers — key targets for retention campaigns.
- Hanoi, Phnom Penh, and Krabi emerged as top-revenue travel packages, indicating demand for premium experiences.
- Discounts are widely used (applied in 60%+ of bookings), proving effective in boosting booking volume without significantly harming revenue.
- Customer demographics are balanced by gender and age, with the 35–49 group being the most active travellers.
- Repeat customers make up 42.3%, showing potential for loyalty growth.
- Feedback analysis showed Yangon as the highest-rated destination, while Bagan and Dili had higher refund or lower satisfaction rates.
- Guide performance varied, with a few guides needing attention due to low ratings and higher refund rates.
- Refund rate among dissatisfied customers was 0%, suggesting potential disconnect between dissatisfaction and refund policy awareness.
- Insights will directly power Power BI dashboards, enabling data-driven tracking of KPIs, customer trends, and operational performance.